- Both Winners Said that the data Quality Matters more than the Model complexity

Step ①: Data Filtering — Some Corrupted data are present that (with the training)

↓

Step ② Load Raw Keypoints → 42 hand landmarks (21x2 (L,R))

↓

Step ③ Filter Empty Frames → Winners insight if the lip = 0 in >50% of the frames then drop the whole video

↓

Step ④ Interpolate missing landmarks →

↓

Step ⑤: Anchor Normalization → the signs are not same to all people, this helps it become people independent

↓

Helps in Generalization

Step ⑥ Temporal Resampling → Around 200 → 50 frames long (min-max)

Sub sample them to 64 frames

↓

Step ⑦ Data Augmentation → Making Models Robust to Variations only on the Train set

→ Augmentation doubled Validation Accuracy

↓

⑧ I/p shape: (Batch size, 64, 42, 3) → (x, y, z) cordinates

64 → vid frame length

42 → No. Key points

O/p shape → (Batch-size, 250)

~> Lot of class imbalance Present in the dataset

Used mainly
① Focal loss
② Weighted Random Sampler

→ Sample Rare classes more in training

⑨ Slow training

↓

loading parquet files are too slow so convert them to npz or npy files say to load reduces train time.

⑩ GPU Memory Management

→ ① Load the entire npz on the VRAM.

→ ② Used mix Precision: ½'s memory.

→ ③ Trade compute for Memory.